Unit: Unit 0: Foundations of Algebra
Topic: Variables and Expressions / Order of Operations

Name: ____________________
Date: ____________

Quiz (Carolina Reaper)

Q1. A shipping company has $3 \cdot x + 2$ trucks on the road, where x is the number of delivery routes. If $x = 4$, how many trucks are on the road?
(A) 10
(B) 12
(C) 14
(D) 16
(E) 18

Q2. Evaluate the expression $2^3 + 12 \div 4 - 5$.
(A) 3
(B) 5
(C) 7
(D) 9
(E) 11

Q3. A traffic flow model is given by the expression $x^2 - 4x + 4$. Simplify the expression.
(A) $x^2 - 4x$
(B) $x^2 + 4$
(C) $x^2 - 4x + 4$
(D) $x^2 - 2x$
(E) $x^2 + 2x$

Q4. The time it takes to deliver a package is given by $t = 2x + 5$, where t is the time in hours and x is the distance in miles. If $x = 10$, what is the delivery time?
(A) 15 hours
(B) 20 hours
(C) 25 hours
(D) 30 hours
(E) 35 hours

Q5. Evaluate the expression $(3 + 2) \cdot (8 - 3)$.
(A) 20
(B) 25
(C) 30
(D) 35
(E) 40

Q6. A logistics company has a cost function given by $C = 2x^2 + 5x + 10$, where C is the cost and x is the number of shipments. If $x = 3$, what is the total cost?
(A) 37
(A) 43
(A) 49
(A) 55
(A) 61

Q7. Simplify the expression $3(2x + 1) - 2(x - 3)$.
(A) $4x + 7$
(B) $5x - 2$
(C) $6x + 1$
(D) $7x - 3$
(E) $8x + 2$

Q8. Evaluate the expression $(9 - 3) \div (2 + 1)$.
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Open Ended Questions (FRQ)

Q9. A transportation company has a route that takes $t = x^2 - 4x + 5$ hours to complete, where t is the time and x is the distance in miles. If the company wants to deliver packages to two locations, one at $x = 5$ miles and the other at $x = 10$ miles, what is the total time taken for both deliveries? Show your work and calculate the total time.

Q10. A shipping container has a volume given by $V = 2x^2 + 5x - 3$, where V is the volume in cubic feet and x is the length in feet. If the container is 8 feet long, what is the volume of the container? Show your work and calculate the volume.

Chef's Tips

- Remind students to follow the order of operations (PEMDAS) when evaluating expressions.
- Encourage students to simplify expressions by combining like terms.
- Emphasize the importance of substituting given values into expressions and evaluating results.